

BERT Shows Garden Path Effects
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INTRODUCTION

?? The horse raced past the barn fell. ??

- Sequential processing leads to **temporary ambiguity**
- Syntactic parsing and semantic role assignment** is more complicated because of the ambiguity
- Unidirectional language models** might be superior to humans at garden path processing because they process all tokens simultaneously

QUESTION ANSWERING

Garden Path
The child who was bought the ice cream cone hugged the nanny.

Local Coherence
The nanny hugged the child who was bought the ice cream cone.

BERT Experiment

- 100 contexts x 5 question types = 500 context + question pairs
- The trained models from BERT family, fine-tuned for QA using SQuAD2.0

Behavioral Experiment

- 74 English speakers recruited from ProProfs
- 40 random items, 30 target = 70 filter
- > 80 responses for each of the sentence and question types

REFERENCES

1. Wilson, K., Irwin, T., & Marantz, A. (2020). BERT Shows Garden Path Effects. *Neuroscience of Language*.
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3. Wilson, K., Irwin, T., & Marantz, A. (2020). BERT Shows Garden Path Effects. *Neuroscience of Language*.
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9. Wilson, K., Irwin, T., & Marantz, A. (2020). BERT Shows Garden Path Effects. *Neuroscience of Language*.
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OBJECTIVES

Do humans and BERT perform differently in answering questions about garden path sentences given their distinct processing strategies?

Does BERT also struggle to assign semantic roles in garden path sentences, akin to humans?

PROBE TASK

Probe Training
We trained binary linear classifiers on our model's word embeddings across all layers to probe representations of the **Agent** and **Goal** roles, using annotated sentences from ProProfs Q&A.

- Training set of 5000 token embeddings with a 50/50 split between having or not having the role in question
- 80/20 train-test split
- High accuracy on ProProfs test sentences

Model	Layer	Agent Accuracy	Goal Accuracy
BERT-base	9	84.52	75.95
BERT-large	17	84.44	75.92
RoBERTa-base	11	44.2	43.6
RoBERTa-large	18	64.0	66.44

Probe Results on ProProfs sentences for layers with highest accuracy and selectivity

Probing Garden Paths

- Filtered only sentences containing both Agent and Goal roles, according to ProProfs (37163 sentences total)
- Images represent semantic roles

- The BERT models fail to assign the Goal role** - the head of the embedded clause
- RoBERTa models classify Goal** more frequently
- Size doesn't matter** - Larger models do not perform better universally
- Role classifiers perform very differently** between BERT and RoBERTa models despite identical probe hyperparameters and training set

CONCLUSION

- BERT and RoBERTa models perform similarly on question answering tasks, regardless of size
- BERT family models are worse than humans on garden path sentences despite similar input
- These results may be due to a failure to assign the correct semantic role in relative clauses following theories from human sentence processing
- The models' are inconsistent in which roles are misassigned

Check out our paper and code!

Neuroscience of Language Lab
Supported by NYU Abu Dhabi Institute (Grant 01501)

Hello everyone, thanks for stopping by my virtual presentation. My name is Kyra Wilson, and I'm going to be talking about garden path sentences and language models ability to extract information from them in comparison to humans.

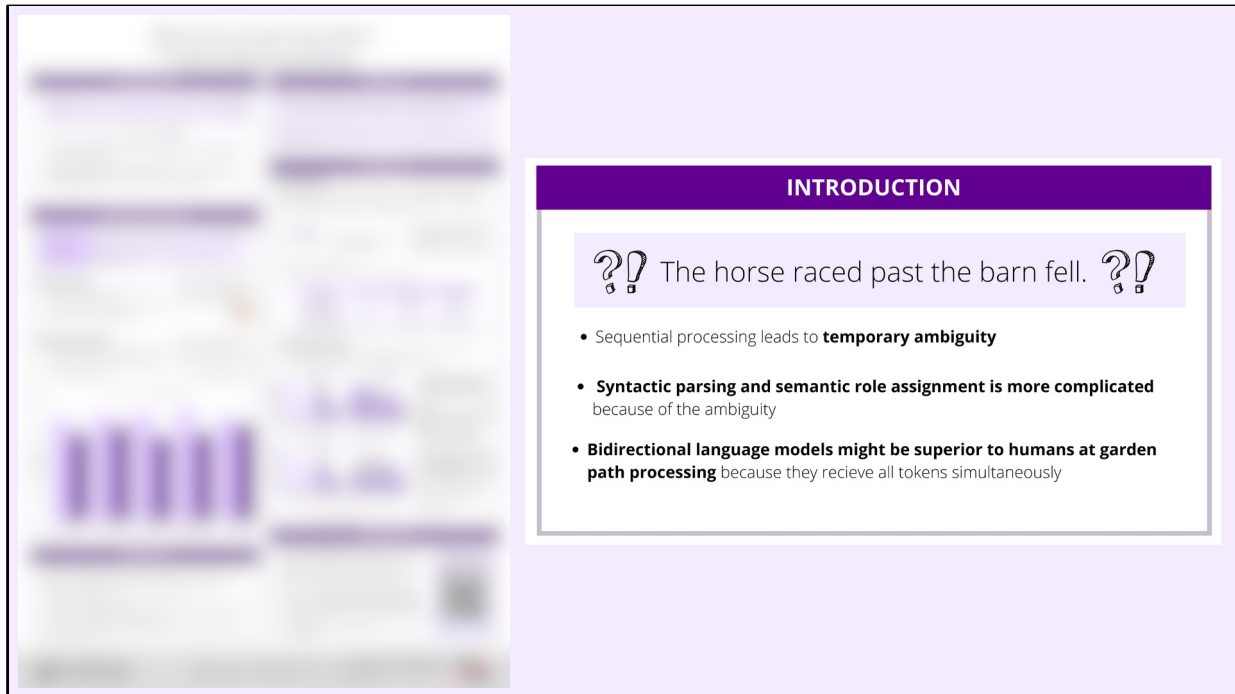
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Now to begin, I'll explain what a garden path sentence actually is. An example of this kind of structure is something like "The Horse Raced Past the Barn Fell". This sentence contains temporary ambiguity as to the correct syntactic parse of the sentence. In our example sentence, the horse is

assumed to be the agent of “raced”, until we arrive at the disambiguating information, in this case, the verb “fell”.

Human readers tend to comprehend garden path sentences poorly. Explanations for this failure to comprehend are varied. On the one hand, it could be that readers fail to construct the correct syntactic structure of the garden path sentence, even after receiving the disambiguating information. Another explanation is that while readers do achieve the correct syntactic parse, the roles assigned to the initial parse stick around, causing interference in comprehension.

These struggles highlight the largely linear nature of human sentence processing. However, other language systems, like BERT, process their input alinearly. This leads to our study’s main questions (next slide)



INTRODUCTION

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- **Syntactic parsing and semantic role assignment is more complicated** because of the ambiguity
- **Bidirectional language models might be superior to humans at garden path processing** because they receive all tokens simultaneously

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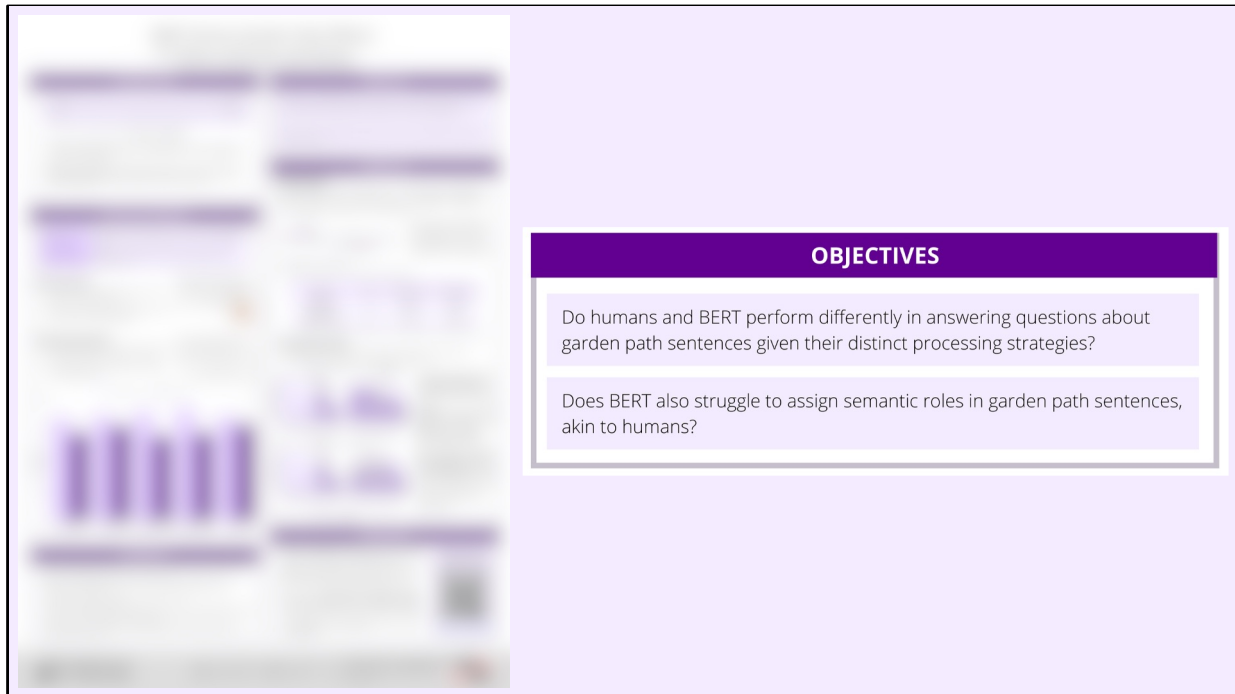
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These struggles highlight the difficulties presented by the largely linear nature of human sentence processing. However, other language systems, like BERT, process their input alinearly. This leads to our study's main questions (next slide)



OBJECTIVES

- Do humans and BERT perform differently in answering questions about garden path sentences given their distinct processing strategies?
- Does BERT also struggle to assign semantic roles in garden path sentences, akin to humans?

First, (READ QUESTION 1). Additionally, since humans may have difficulty processing garden path sentences due to semantic role mis-assignment, is this mis-assignment also present in BERT's embeddings?



QUESTION ANSWERING

Garden Path	The child (who was) bought the ice cream cone hugged the nanny.
Local Coherence	The nanny hugged the child (who was) bought the ice cream cone.

BERT Experiment

- 160 contexts x 5 question types = 800 context + question pairs
- Pre-trained models from BERT family, fine-tuned for QA using SQuAD2.0

Behavioral Experiment

- 74 English speakers recruited from Prolific
- 45 random items: 32 target + 13 filler
- > 80 responses for each of the sentence and question types

Context: The nanny hugged the child bought the ice cream cone.
Question: Who hugged the child?



The nanny hugged the child bought the ice cream cone.

Who hugged the child?



To address the first question, we designed a question answering task to compare the performance of humans and BERT. In this task, humans or BERT would be presented with a garden path (or similar) sentence, and then be asked a question aimed to assess their comprehension.

Our garden path sentences were based on a set used in a previous experiment, so they also included another kind of structure similar to garden paths termed “local coherence”.

The difference between the two is that where garden paths have an embedded clause in the middle of the sentence that leads to a temporary ambiguity, while in local coherence sentences the embedded clause falls at the end of the sentence so it lacks ambiguity and should be easier to understand.

The benefit of including both kinds of sentences is that we are able to analyze whether it is simply embedded sentences which make processing difficult or whether the structural positioning of those sentences also plays a role.

For the BERT portion, we had 160 garden path or local coherence structures and five comprehension questions for each, which led to a total of 800 test items.

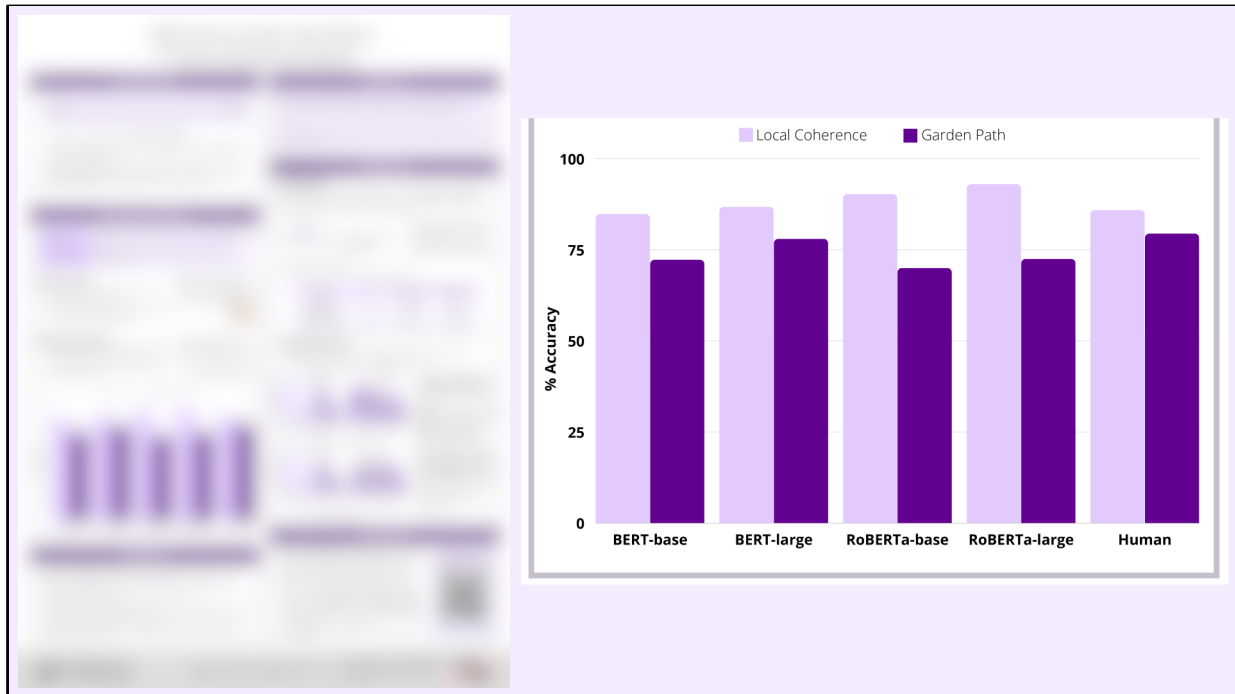
We gave the full set to 4 different models in the BERT family:

BERT base and large, as well as RoBERTa base and large.

Each of the models had been fine-tuned to perform question answering using the SQuAD 2.0 dataset.

For the behavioral portion, we used Prolific to recruit 74 English speakers.

They each saw 32 items which had been randomly selected from the 800 test items, in addition to 13 filler items. This resulted in at least 80 responses for each of the sentences, which we scored and averaged to compare to the results from the BERT models.



As you can see in the plot here, across all models and humans, performance on LCS is above that of GPS. This suggests that there is something unique about the structure of the GPS and the resulting ambiguity does make language processing more difficult across all systems.

Additionally, while the language models'

performance is generally above that of humans on LCS, they perform worse across the board on GPS. This suggests that even though these LMs process input a linearly, they still face difficulties in forming a correct parse or assigning semantic roles, just as humans do.

Finally, we note that there is not one singular model which performs better or worse than the others. While RoBERTa-large has the highest performance on LCS, it also has the one of the worst performances on GPS. Similarly, BERT-base has the worst performance on LCS, but its performance on GPs is on par with two of the three other models.

In the next section of this investigation, we examine this further by looking at the internal representations in each of the models to both determine what is causing the difficulties in processing GPS and whether any similarities or differences exist between the models that could explain their downstream performance.



PROBE TASK

Probe Training

We trained binary linear classifiers on our model's word embeddings across all layers to probe representations of the **Agent** and **Goal** roles, using annotated sentences from PropBank [5].



- Training set of 5000 token embeddings with a 50/50 split between having or not having the role in question.

- 80/20 train test split
- High accuracy on Propbank test sentences

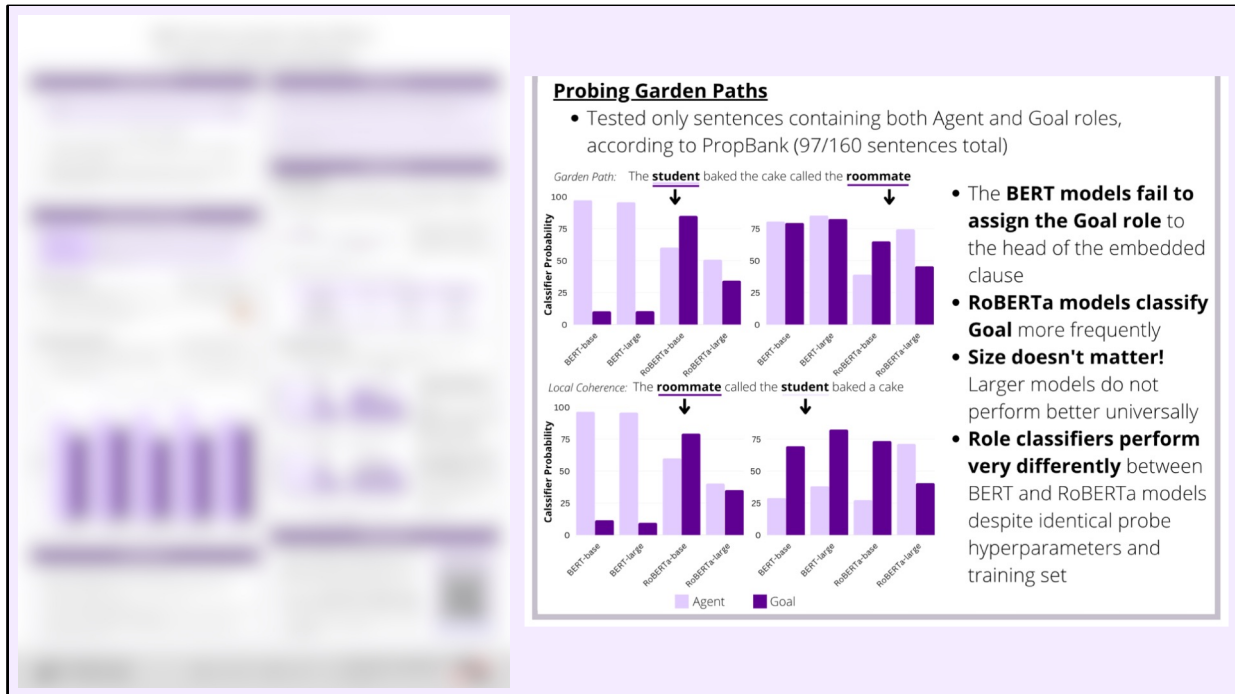
Model	Layer	Agent Accuracy	Goal Accuracy
BERT-base	9	84.92	75.96
BERT-large	17	84.44	75.12
RoBERTa-base	11	64.2	62.6
RoBERTa-Large	10	66.8	66.44

Probe Results on Propbank sentences for layers with highest accuracy and selectivity

To answer these questions,
 we performed a probe task by training a linear classifier
 on non-garden path sentences which had been
 annotated for semantic roles in PropBank.
 We focused specifically on the agent and goal roles,
 because those are the ones which are essential for
 interpreting garden path sentences.
 We tested the classifiers ability to accurately identify
 the semantic role
 within the embedded representations of a given
 token

for all layers in all models,

and then selected the layer from each model with the highest accuracy for investigating the garden path sentences.



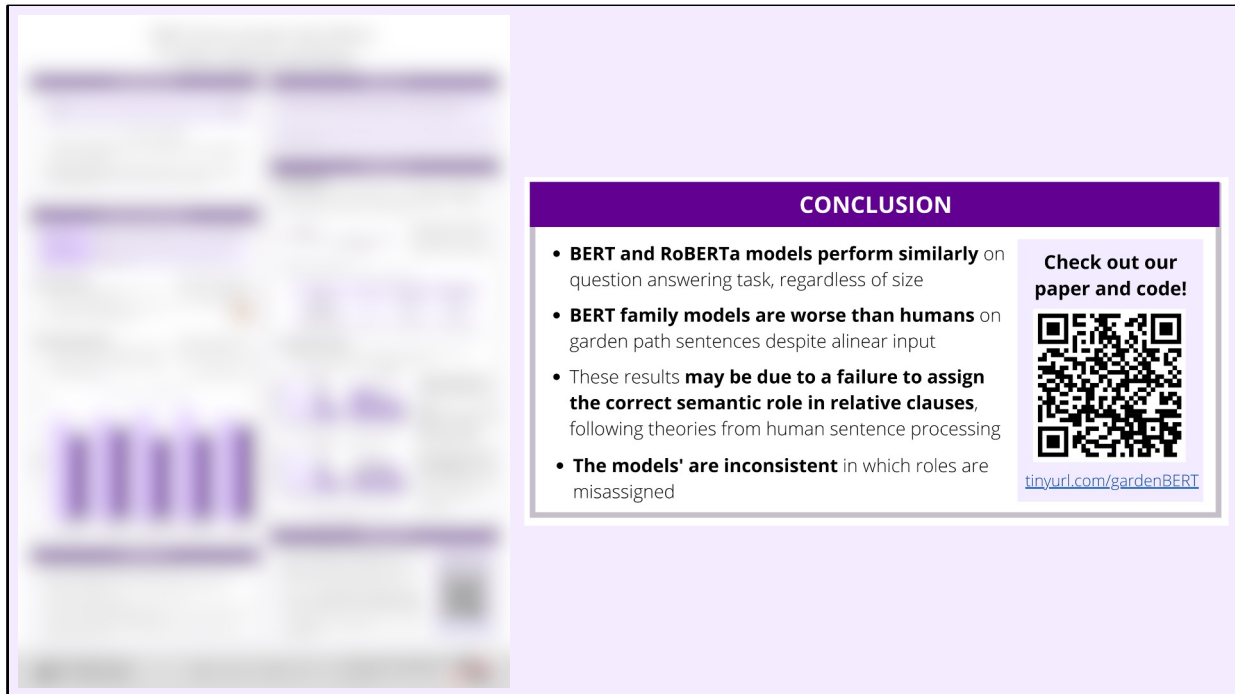
Because of inconsistencies in the PropBank annotations,
we chose to only analyze the sentences which had an Agent and Goal role,
according to PropBank.
This left us with a total of 97 out of 160 sentences from the question answering dataset.

For BERT models, we found that the Goal role was not strongly represented on the head of the embedded clause in the GPS. In contrast, it was represented in the local coherence sentences. However, it is not clear whether BERT was simply representing the goal role from the main clause, the embedded clause, or both in these sentences.

RoBERTa-base, on the other hand, does show higher probabilities of classifying something as a goal. However, it seems to universally prefer goal role assignment over agent roles, as evidenced by low agent probabilities on all arguments. RoBERTa-large patterns in the reverse, and generally seems to prefer assigning agent roles over goal roles for all entities.

The final important result from this probe task is that increasing model size does not mean better representations of semantic roles— for BERT, the most likely role does not change when adding parameters,

and for RoBERTa, there is a change in prediction but the larger model is neither more right nor wrong than the smaller one.



CONCLUSION

- **BERT and RoBERTa models perform similarly** on question answering task, regardless of size
- **BERT family models are worse than humans** on garden path sentences despite a linear input
- These results **may be due to a failure to assign the correct semantic role in relative clauses**, following theories from human sentence processing
- **The models' are inconsistent** in which roles are misassigned

Check out our paper and code!



tinyurl.com/gardenBERT

Finally, with the results from the question answering task and the probe task,

we are ready to circle back to the two questions I initially proposed.

First, given that humans and BERT-style models receive

their inputs in very different forms,
how does this affect their ability to answer questions about GPS?

In the first part of this study we compared the two in a QA task

and found that the performance of BERT and

RoBERTa is quite similar,
and no model outperforms all the others.
Secondly, we observed that models in the BERT
family actually perform worse
than humans on GPS despite not being subjected
to the linear presentation which causes the
ambiguity.

Second, now that we know that BERT also suffers
from GP effects,
what is the cause of this?
Using a probe task, we determined that both BERT
and RoBERTa's representations
of the semantic roles in garden path sentences are
incomplete.
This misrepresentation of semantic roles is also a
leading theory
of why humans have difficulties understanding
garden path sentences.

Another interesting result from the probe task was that the semantic representations are not consistent between models, and in the case of RoBERTa, between sizes of models.

This leads to some very interesting directions of further research, including how the models' different training procedures and sizes can result in very disparate internal representations, and how downstream performance in a QA task can still be relatively similar despite those representational differences.

I'd like to conclude by thanking you for your time, and If you've enjoyed this presentation, please check out our code and paper at the link here!

We have more discussion about the kinds of questions we developed and used for the question answering task, as well as an additional manipulation of disambiguation that we did not have space to discuss here. Also feel free to reach out if you have any questions or comments, and thanks again for watching!